

HDOC 22 ASSIGNMENT

Aim: To write a menu-driven program in C using loops and switch statements to read two positive integers m and n ($m \leq n$) and perform operations based on user selection: side-by-side multiplication tables, a multiplication grid, or printing consecutive integers, repeating until the user exits.

Algorithm:

1. Start the program.
2. Declare integer variables: m, n, choice, i, and j.
3. Input two positive integers m and n. If $m > n$ or if either is less than 1, display an error message and ask again.
4. Start a do-while loop for the menu:
 - a. Step 4.1: Display menu options (1. Tables, 2. Grid, 3. Consecutive Integers, 4. Exit).
 - b. Step 4.2: Read user choice.
 - c. Step 4.3: Execute using switch(choice):
 - i. **Case 1:** Run a loop i from 1 to 10. Print $m * i$ and $n * i$ side by side.
 - ii. **Case 2:** Print top numbers from 1 to n. Use nested loops (i from 1 to m and j from 1 to n) to print the grid product $i * j$.
 - iii. **Case 3:** Run a outer loop i from m to n. Inside, run a loop j from i to i + 5 to print 6 numbers total for each integer.
 - iv. **Case 4:** Print exit message.
 - v. **Default:** Print "Invalid choice! Please try again."
5. Repeat the loop until choice == 4.
6. End the program.

C Program Code:

```

#include <stdio.h>

int main() { int m, n;int choice;int i, j;

// Input values for m and n
printf("Enter positive integer m: ");
scanf("%d", &m);
printf("Enter positive integer n (n >= m): ");
scanf("%d", &n);

// Basic check for valid inputs
if (m <= 0 || n <= 0 || m > n) {
    printf("Invalid input! Ensure m <= n and both are positive numbers.\n");
    return 0;
}

do {
    // Display Menu
    printf("\n1. Multiplication tables of m and n\n");
    printf("2. Multiplication grid\n");
    printf("3. Consecutive integers\n");
    printf("4. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice) {
        case 1:
            printf("\nMultiplication Tables of %d and %d (m=%d, n=%d)\n", m, n, m, n);
            for(i = 1; i <= 10; i++) {
                printf("%d x %2d = %-4d\t%d x %2d = %-4d\n", m, i, m * i, n, i, n * i);
            }
            break;

        case 2:
            printf("\nMultiplication Grid (m=%d, n=%d)\n", m, n);

```

```

    printf(" ");
    for(j = 1; j <= n; j++) {
        printf("%3d ", j);
    }
    printf("\n-----\n");

    for(i = 1; i <= m; i++) {
        printf("%2d |", i);
        for(j = 1; j <= n; j++) {
            printf("%3d ", i * j);
        }
        printf("\n");
    }
    break;

case 3:
    printf("\nConsecutive Integers from %d to %d (m=%d, n=%d)\n", m, n, m, n);
    for(i = m; i <= n; i++) {
        printf("%d: ", i);
        for(j = i; j <= i + 5; j++) {
            printf("%d ", j);
        }
        printf("\n");
    }
    break;

case 4:
    printf("Exiting program...\n");
    break;

default:
    printf("Invalid choice. Please enter a number between 1 and 4.\n");
}
} while(choice != 4);

return 0;

}

```

Code screenshot :

```
#include <stdio.h>

int main() {
    int m, n;
    int choice;
    int i, j;

    // Input values for m and n
    printf("Enter positive integer m: ");
    scanf("%d", &m);
    printf("Enter positive integer n (n >= m): ");
    scanf("%d", &n);

    // Basic check for valid inputs
    if (m <= 0 || n <= 0 || m > n) {
        printf("Invalid input! Ensure m <= n and both are positive numbers.\n");
        return 0;
    }

    do {
        // Display Menu
        printf("\n1. Multiplication tables of m and n\n");
        printf("2. Multiplication grid\n");
        printf("3. Consecutive integers\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice) {
            case 1:
                printf("\nMultiplication Tables of %d and %d (m=%d, n=%d)\n", m, n, m, n);
                for(i = 1; i <= 10; i++) {
                    printf("%d x %2d = %-4d\t%d x %2d = %-4d\n", m, i, m * i, n, i, n * i);
                }
                break;

            case 2:
                printf("\nMultiplication Grid (m=%d, n=%d)\n", m, n);
                printf(" ");
                for(j = 1; j <= n; j++) {
                    printf("%3d ", j);
                }
                printf("\n-----\n");

                for(i = 1; i <= m; i++) {
                    printf("%2d |", i);
                    for(j = 1; j <= n; j++) {
                        printf("%3d ", i * j);
                    }
                    printf("\n");
                }
                break;
        }
    } while(choice != 4);
}
```

```
    case 3:
        printf("\nConsecutive Integers from %d to %d (m=%d, n=%d)\n", m, n, m, n);
        for(i = m; i <= n; i++) {
            printf("%d: ", i);
            for(j = i; j <= i + 5; j++) {
                printf("%d ", j);
            }
            printf("\n");
        }
        break;

    case 4:
        printf("Exiting program...\n");
        break;

    default:
        printf("Invalid choice. Please enter a number between 1 and 4.\n");
}
} while(choice != 4);

return 0;
}
```

Output screenshot :

```

[manalithakar@Manalis-MacBook-Air ~ % nano 22q1.c
[manalithakar@Manalis-MacBook-Air ~ % gcc 22q1.c -o 22q1
[manalithakar@Manalis-MacBook-Air ~ % ./22q1
Enter positive integer m: 2
Enter positive integer n (n >= m): 4

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1

Multiplication Tables of 2 and 4 (m=2, n=4)
2 x 1 = 2      4 x 1 = 4
2 x 2 = 4      4 x 2 = 8
2 x 3 = 6      4 x 3 = 12
2 x 4 = 8      4 x 4 = 16
2 x 5 = 10     4 x 5 = 20
2 x 6 = 12     4 x 6 = 24
2 x 7 = 14     4 x 7 = 28
2 x 8 = 16     4 x 8 = 32
2 x 9 = 18     4 x 9 = 36
2 x 10 = 20    4 x 10 = 40

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2

Multiplication Grid (m=2, n=4)
      1  2  3  4
-----
1 | 1  2  3  4
2 | 2  4  6  8

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 3

Consecutive Integers from 2 to 4 (m=2, n=4)
2: 2 3 4 5 6 7
3: 3 4 5 6 7 8
4: 4 5 6 7 8 9

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 4
Exiting program...
manalithakar@Manalis-MacBook-Air ~ %

```